

Spectral Decomposition Based Trading Strategy

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1. Data Setup

Let:

- $N = 500$ (number of stocks)
- T (number of time steps)
- $P_i(t)$: price of stock i at time t

Log return:

$$R_i(t) = \log P_i(t) - \log P_i(t - 1)$$

Log return matrix $\mathbf{R} \in \mathbb{R}^{(T-1) \times N}$:

$$\mathbf{R} = [R_i(t)]_{t=2}^T$$

2. Correlation Graph

Correlation matrix **C**:

$$C_{ij} = \text{Corr}(R_i, R_j)$$

Graph $G = (V, E)$:

- Nodes: $V = \{1, \dots, N\}$
- Edge weights: $w_{ij} = C_{ij}$ or $\exp\left(-\frac{1-C_{ij}}{\sigma^2}\right)$

Adjacency matrix **W**, Degree matrix **D**:

$$D_{ii} = \sum_j w_{ij}, \quad \mathbf{L} = \mathbf{D} - \mathbf{W}$$

Normalized Laplacian:

$$\mathcal{L} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}$$

3. Spectral Clustering

Compute smallest k eigenvectors of $\mathcal{L}$:

$$\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_k] \in \mathbb{R}^{N \times k}$$

Normalize rows:

$$\tilde{\mathbf{u}}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}$$

Apply k-means to rows $\tilde{\mathbf{u}}_i$:

- Assign stock i to cluster $c(i) \in \{1, \dots, k\}$

4. Price Normalization

Normalized prices:

$$\tilde{P}_i(t) = \frac{P_i(t) - \mu_i}{\sigma_i}, \quad \mu_i = \frac{1}{T} \sum_t P_i(t)$$

Each stock in cluster c :

- $S_c = \{i : c(i) = c\}$

Centroid series:

$$\bar{P}_c(t) = \frac{1}{|S_c|} \sum_{i \in S_c} \tilde{P}_i(t)$$

5. Signal Generation

Cluster spread:

$$\sigma_c(t) = \sqrt{\frac{1}{|S_c|} \sum_{i \in S_c} (\tilde{P}_i(t) - \bar{P}_c(t))^2}$$

Z-score:

$$z_i(t) = \frac{\tilde{P}_i(t) - \bar{P}_c(t)}{\sigma_c(t)}$$

Trading Signal:

- $z_i(t) > 3$: Short stock i
- $z_i(t) < -3$: Buy stock i
- Else: Hold

Optional Enhancements

- Add transaction cost buffer: $|z_i(t)| > 3 + \delta$
- Use rolling window normalization
- Exponential volatility adjustment
- Integrate signal with portfolio optimizer